

Labortory 2

Spatial Filtering and Thresholding

ECE 7410: Image Processing

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Spatial Filtering

Smoothing

1. Download the test images (*img1.png* and *img2.png*) from Brightspace under Lab 2.



2. Read *img1.png*, convert it to a grayscale image, and display.

See the following image *img1_grey.png*

3. Develop and display a function that will perform spatial filtering (i.e., convolution) on an $M \times N$ pixel image using a general $m \times n$ kernel. The function should take an image and a kernel as inputs and return the filtered image. Do **NOT** use internal functions for this (e.g., MATLAB's *imfilter* function). (*hint*: zero padding the edges of the image is required).

4. Filter *img1.png* with each of the five filters below (Kernel 1 to Kernel 5) using your function and display the outputs:

1. $\text{Kernel_1} = \left(\frac{1}{9}\right) * \text{ones}(0)$

See Figure 1 below.

img1.png with Kernel_1 applied

Figure 1: *img1.png* with Kernel_1 applied

2. $\text{Kernel_2} = \left(\frac{1}{49}\right) * \text{ones}(7)$

See Figure 2 below.

img1.png with Kernel_2 applied

Figure 2: *img1.png* with Kernel_2 applied

3. $\text{Kernel_3} = \text{fspecial}(\text{'average'}, [7, 7])$

See Figure 3 below.

img1.png with Kernel_3 applied

Figure 3: *img1.png* with Kernel_3 applied

4. $\text{Kernel_4} = \text{fspecial}(\text{'gaussian'}, [3, 3], 0.5)$

See Figure 4 below.

5. $\text{Kernel_5} = \text{fspecial}(\text{'gaussian'}, [7, 7], 1.2)$

See Figure 5 below.

5. Develop and display a function that performs median filtering. A median filter will replace the pixel intensity with the median intensity of the pixels overlapped by the kernel. The function should take the image and the size of the kernel as inputs. (*hint*: zero padding the edges of the image is required).

img1.png with Kernel_4 applied

Figure 4: img1.png with Kernel_4 applied

img1.png with Kernel_5 applied

Figure 5: img1.png with Kernel_5 applied

6. Apply the following two median filters to *img1.png* and display the outputs:

1. Median filter with kernel size 3×3

See Figure 6 below.

img1.png with median filter 3×3 applied

Figure 6: img1.png with median filter 3×3 applied

2. Median filter with kernel size 5×5

See Figure 7 below.

7. Compare and discuss the smoothing effects obtained by *Kernel_1* to *Kernel_5* and the two median filters.

...

8. Discuss the effect the smoothing kernel size has on the output images in your experiments.

...

Sharpening

1. Use the function you developed in **Smoothing: Step 3** and apply the following three sharpening filters (*Kernel_6* to *Kernel_8*) to *img2.png*.

$$1. \text{Kernel}_6 = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & -2 \\ -1 & 0 & 1 \end{bmatrix}$$

See Figure 8 below.

$$2. \text{Kernel}_7 = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

See Figure 9 below.

$$3. \text{Kernel}_8 = \text{fspecial('log', 3)}$$

See Figure 10 below.

2. Compare and discuss the outputs obtained by *Kernel_6* to *Kernel_8*

...

img1.png with median filter 5×5 applied

Figure 7: img1.png with median filter 5×5 applied

img2.png with Kernel_6 applied

Figure 8: img2.png with Kernel_6 applied

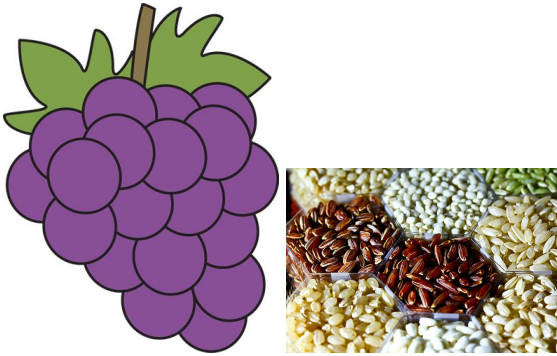
Thresholding

The following equations are from the lab manual.

$$\omega_0(t) = \sum_{i=0}^{t-1} p(i) \quad \mu_0(t) = \sum_{i=0}^{t-1} ip(i)/\omega_0 \quad \omega_1(t) = \sum_{i=t}^{L-1} p(i) \quad \mu_1(t) = \sum_{i=t}^{L-1} ip(i)/\omega_1 \quad (1)$$

$$\sigma_b^2(t) = \omega_1 \omega_2 [\mu_0 - \mu_1]^2 \quad (2)$$

1. Download the test images (*img3.png* and *img4.png*) from Brightspace under Lab 2 and save them in your working directory.



2. Develop and display code to calculate the class probabilities and class means, as well as the inter-class variance (equations 1 and 2) as a complete function to calculate the optimal threshold value for any given image.
3. Calculate and report the optimal threshold values for *img3.png* and *img4.png* found using your code in **Thresholding: Step 2**.
 - *img3.png*:
— ...
 - *img4.png*:
— ...
4. Develop code to convert a grayscale image to a binary image using a given threshold value. Do **NOT** use internal functions for this (e.g., the MATLAB function *im2bw*).
5. Use the optimal thresholds to generate and display binary image for *img3.png* and *img4.png*.
See Figure 11 and Figure 12 below.

img2.png with Kernel_7 applied

Figure 9: img2.png with Kernel_7 applied

`img2.png` with `Kernel_8` applied

Figure 10: `img2.png` with `Kernel_8` applied

binary image of `img3.png` with *threshold* = `XXX`

Figure 11: binary image of `img3.png` with *threshold* = `XXX`

binary image of `img4.png` with *threshold* = `XXX`

Figure 12: binary image of `img4.png` with *threshold* = `XXX`